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SPECIAL ARTICLE

# Using ChatGPT to perform a systematic review: a tutorial

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## ABSTRACT

This tutorial provides a comprehensive guide on leveraging ChatGPT for systematic literature reviews, leveraging actual applications in cardiovascular research. Systematic reviews, while essential, are resource-intensive, and ChatGPT offers a potential solution to streamline the process. The tutorial covers the entire review process, from preparation to finalization. In the preparation phase, ChatGPT assists in defining research questions and generating search strings. During the screening phase, ChatGPT can efficiently screen titles and abstracts, processing multiple abstracts simultaneously. The tutorial also introduces an intermediate step of generating study summaries that leads to the generation of reliable data extraction tables. For assessing the risk of bias, ChatGPT can be prompted to perform these tasks. Using each tool's explanation document to generate an appropriate prompt is an efficient method of reliable risk of bias assessments using ChatGPT. However, users are cautioned about potential hallucinations in ChatGPT's outputs and the importance of manual validation. The tutorial emphasizes the need for vigilance, continuous refinement, and gaining experience with ChatGPT to ensure accurate and reliable results. The methods presented have been successfully tried in several projects, but they remain in nascent stages, with ample room for improvement and refinement.

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KEY WORDS: Artificial intelligence; Systematic review; Machine learning.

The incorporation of artificial intelligence (AI) and machine learning (ML) into medical practices heralds a transformative era for healthcare, with cardiology being no exception. The promise of precision cardiovascular medicine, achieved through the synergies of AI and ML, is taking shape and offering groundbreaking insights.[1] Particularly, deep learning, a subset of ML, has demonstrated immense potential, notably in cardiac contractile dysfunction detection through AI-enabled electrocardiograms.[2]

In the realm of predictive analytics within cardiology, ML techniques are already bearing fruit. The ability to predict events, such as 30-day all-cause readmissions in heart failure patients, is advancing, thanks to these sophisticated computational approaches.[3] These advancements are not just isolated to direct patient care. The integration of big data analytics in healthcare offers unparalleled insights, where vast amounts of data are processed, analyzed, and distilled into actionable knowledge, thereby propelling healthcare towards new frontiers.[4]

Historically, systematic reviews have stood as the bedrock of evidence-based medicine. Yet, as

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invaluable as they are, their comprehensive nature often demands extensive time and resources. The ever-expanding universe of medical literature only intensifies this challenge. Enter AI: offering a promise to revolutionize this space by streamlining the systematic review process. The use of text mining for study identification in systematic reviews, as an example, has proven to be a robust application of AI, refining the accuracy and efficiency of the process.[5]

It is paramount, then, for cardiologists to remain conversant with these groundbreaking technologies. The aim of this tutorial is not only to familiarize cardiologists with the theoretical underpinnings but also to provide hands-on guidance on how to leverage ChatGPT in conducting systematic reviews. The combination of AI's prowess and the specific nuances of cardiology promises to usher in an age of enhanced medical research, ultimately ensuring superior patient outcomes.

## Basic concepts

Before we begin to dwell on specific techniques for performing a systematic review with ChatGPT, it is imperative that we understand some of the basic mechanics of how it operates. The text that follows is aimed at introducing some of the fundamental concepts of how this tool works. It is highly recommended that further reading on these topics is pursued. Furthermore, it should be noted that the methodology we present is based on ChatGPT version 4.0 (paid version). We have not tried to apply this method to ChatGPT version 3.5 (free version), and therefore we cannot recommend this methodology is applied to the free version of the model.

### Tokens: the linguistic units of ChatGPT-4.0

Within the landscape of artificial intelligence and natural language processing, tokens stand as essential units of information. In ChatGPT-4.0, a token can signify a single character or span an entire word. An illustrative example would be the statement, "artificial intelligence holds promise." This set of words, when processed, divides into distinct tokens, enabling the model to effectively analyze and craft responses.

*Relevance to systematic review*

A critical point of awareness for users is that ChatGPT-4.0 has a specific token limit. The token limit for a single interaction, *i.e.*, the combined count of both input and output tokens, is likely 2096 (there seems to be debate over this). The total chat history limit is 8192 tokens. This understanding is key when interacting with the model, especially when navigating the detailed corridors of medical literature. OpenAI offers a tokenizer that allows you to check the token count of your interaction(s). It is highly advisable you use that and try not to exceed these limits. The total chat history limit is of particular importance, since, if exceeded, ChatGPT cannot access the entire text for context and may omit vital information.

### Decoding variability: the roles of Temperature and Top P

Embedded within ChatGPT's outputs is a nuanced variability. Although users of the chatbox interface cannot adjust these parameters directly, understanding the roles of Temperature and Top P provides insight. Both are integral in calibrating the randomness and specificity of the model's replies.

*Relevance to systematic review*

Different interactions with ChatGPT-4.0 might yield subtle variances in responses, even with the same query. This characteristic is shaped by underlying mechanisms like Temperature and Top P. Recognizing this variability allows users to extract diverse insights and perspectives from the AI's knowledge reservoir.

Example: in this example, we used the same prompt three times to investigate the variability of the output.

Prompt: please define major adverse cardiovascular outcomes.

*Response 1*

Major Adverse Cardiovascular Outcomes (MACE) typically refer to a composite of significant cardiovascular events that are used as a primary endpoint in clinical trials and observational studies. These outcomes are critical for assessing

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the efficacy of cardiovascular treatments and the cardiovascular risk in patients. MACE includes, but may not be limited to:

• cardiovascular death: death from cardiovascular causes, such as heart attack or stroke;

• myocardial infarction (MI): commonly known as a heart attack, it occurs when blood flow to a part of the heart is blocked for a long enough time that part of the heart muscle is damaged or dies;

• stroke: occurs when the blood supply to part of the brain is interrupted or reduced, preventing brain tissue from getting oxygen and nutrients;

• hospitalization for unstable angina: a condition in which the heart does not get enough blood flow and oxygen, leading to a risk of heart attack, often requiring hospitalization for treatment;

• coronary revascularization: procedures such as angioplasty or coronary artery bypass surgery that restore or improve blood flow to the heart.

The specific components of MACE can vary slightly depending on the study or clinical trial, with some also including outcomes like hospitalization for heart failure. The use of MACE as an endpoint allows for a more comprehensive assessment of cardiovascular risk or treatment benefit over time, incorporating several significant and clinically relevant outcomes.

*Response 2*

Major adverse cardiovascular outcomes (MACE) is a composite endpoint frequently used in cardiovascular research to measure the effectiveness of treatments or interventions. MACE typically includes a range of severe cardiovascular events that are considered significant for patient health. These events usually encompass:

• myocardial infarction (heart attack): this is a condition where there is a blockage of blood flow to the heart muscle, causing tissue damage or death;

• stroke: a stroke occurs when the blood supply to part of the brain is interrupted or reduced, preventing brain tissue from getting oxygen and nutrients. Strokes can be ischemic (due to blockages) or hemorrhagic (due to bleeding);

• death from cardiovascular causes: this refers to death resulting directly from cardiovascular disease or its complications;

• hospitalization for unstable angina: unstable angina is a condition in which the heart does not receive enough blood flow and oxygen, often indicating a blockage or imminent heart attack. Hospitalization is required for stabilization and treatment;

• coronary revascularization: this includes procedures like coronary artery bypass grafting (CABG) or percutaneous coronary intervention (PCI), which are undertaken to restore or improve blood flow to the heart.

The specific components included in MACE can vary slightly between studies, depending on the research question and the patient population. However, the concept always revolves around significant adverse events related to cardiovascular health, offering a way to quantify the impact of a treatment or disease progression on serious cardiovascular outcomes.

*Response 3*

Major adverse cardiovascular outcomes, often abbreviated as MACE, is a term used in clinical research and healthcare to describe critical negative events related to cardiovascular disease. These outcomes typically include:

• cardiovascular death: this refers to death caused by any cardiovascular issue, such as heart attack or stroke;

• myocardial infarction (MI): commonly known as a heart attack, MI occurs when blood flow to a part of the heart is blocked for a long enough time that part of the heart muscle is damaged or dies;

• stroke: a stroke happens when the blood supply to part of the brain is interrupted or reduced, preventing brain tissue from getting oxygen and nutrients. Brain cells begin to die in minutes;

• hospitalization for unstable angina or other acute coronary syndromes: Unstable angina is a condition in which the heart does not receive enough blood flow and oxygen. It may lead to a heart attack and is considered a medical emergency;

• revascularization procedures: these are procedures aimed at restoring blood flow to the heart or other parts of the body, such as coronary artery bypass surgery or percutaneous coronary intervention (angioplasty and stent placement).

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The specific events included in the definition of MACE can vary slightly depending on the study or context, but the focus is always on serious, life-threatening cardiovascular events that can dramatically impact a person's health and survival. Tracking these outcomes is crucial for evaluating the effectiveness and safety of cardiovascular treatments and interventions.

*Comment*

Consistencies across responses:

- definition of MACE: all responses agree that MACE is a composite endpoint used in cardiovascular research to assess the effectiveness of treatments or interventions, focusing on critical cardiovascular events;
- common components of MACE:
  - myocardial infarction (heart attack): mentioned in all responses with a consistent definition;
  - stroke: consistently defined across all responses;
  - cardiovascular death: all responses include death from cardiovascular causes;
  - hospitalization for unstable angina: mentioned in all responses, highlighting its significance as part of MACE;
  - coronary revascularization: all include procedures to restore or improve blood flow, such as angioplasty or bypass surgery.

Discrepancies among responses:

- specificity in definitions:
  - response 1 provides a more detailed introduction to MACE, emphasizing its role in assessing cardiovascular treatments and risk, which is less explicitly detailed in responses 2 and 3;
  - response 2 and response 3 are more aligned in defining MACE primarily as a measure in research without the initial emphasis on clinical trial endpoints;
- variation in event descriptions:
  - unstable angina inclusion: response 3 mentions "unstable angina or other acute coronary syndromes," suggesting a slightly broader categorization than the specific "hospitalization for unstable angina" mentioned in responses 1 and 2;
  - coronary revascularization: while all responses mention revascularization procedures, the terminology and examples provided (*e.g.*, mentioning both angioplasty and coronary artery bypass surgery) are more detailed in some responses than others.
- additional outcomes:
  - only response 1 hints at the possibility of including additional outcomes like "hospitalization for heart failure" in some studies, indicating a slight variation in the inclusivity of the MACE definition.

*Conclusion*

The discrepancies between the responses are relatively minor and primarily revolve around the level of detail provided and the slight variation in describing the components of MACE. All responses are consistent in the core elements that comprise MACE but differ slightly in how broadly they describe these components and the potential inclusion of additional outcomes. This variation reflects the flexibility in defining MACE in different clinical research contexts, where the specific components may slightly vary based on the study's focus, the patient population, or the intervention being tested.

## Contextual clarity and computational capacity in ChatGPT-4.0

In communication, context is invaluable. For ChatGPT-4.0, this holds true; the model thrives within the immediate context it's provided. Without a memory of previous interactions, every engagement with ChatGPT-4.0 starts anew.

*Relevance to systematic review*

The immense training data behind ChatGPT-4.0 implies a vast depth of information at its disposal. Given the intricate nature of many topics, ensuring the prompt is as defined and detailed as possible becomes crucial. By breaking down intricate concepts into their foundational elements, users can better steer the model's processing resources. This not only optimizes the computational capacity but also fine-tunes the outputs.

For instance, rather than inputting a vague question like "Tell me about cardiovascular conditions?", one might provide a more detailed

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prompt: "Provide a summary of atrial fibrillation, its common causes, and the latest treatment approaches in the last decade." This specificity not only conserves tokens but also directs ChatGPT-4.0's computational focus, yielding a more targeted and informative response.

### Prompt engineering: crafting effective queries

The art of constructing queries, often termed 'prompt engineering,' can significantly influence the quality of outputs. Being clear, concise, and direct can lead to more precise and useful responses from ChatGPT-4.0.

*Relevance to systematic review*

Crafting effective queries becomes particularly essential in a systematic review context, where precision and accuracy are paramount. Cardiologists can refine their queries to extract specific types of information, be it randomized control trial results, meta-analysis data, or patient cohort studies.

### Iterative querying: deepening understanding through repetition

The notion of asking related follow-up questions, or iterating on original queries, can enhance the depth and breadth of information extracted.

*Relevance to systematic review*

Sometimes, a single interaction might not capture the entirety of a topic or provide a sufficiently detailed response. Cardiologists can use iterative querying to dive deeper into specific areas, ensuring a thorough exploration of the literature.

### Bias and neutrality: recognizing inherent model tendencies

No machine learning model, including ChatGPT-4.0, is entirely free from biases. These biases stem from the data the model was trained on.

*Relevance to systematic review*

While ChatGPT-4.0 can be a powerful tool in mining literature and information, cardiologists should approach the data with a critical eye, cross-referencing and being wary of any potential underlying biases in the outputs.

### Model limitations: the importance of human oversight

While ChatGPT-4.0 is incredibly advanced, it is not infallible. Recognizing the areas where human oversight is crucial ensures that the AI's outputs are utilized effectively and safely.

*Relevance to systematic review*

Human expertise, especially in specialized fields like cardiology, remains irreplaceable. Cardiologists should leverage ChatGPT-4.0 as a tool to aid and augment their research processes, but always with a discerning perspective, evaluating the AI-generated content against their clinical knowledge and the broader body of evidence.

By combining the capabilities of ChatGPT-4.0 with these nuanced approaches and understandings, cardiologists can better harness the potential of AI in systematic reviews. This harmonious melding of technology and human expertise promises a future of more streamlined, comprehensive, and insightful research endeavors.

### Commercially available tools

There are several commercial tools that utilize large language models (LLMs) to automate the systematic review process. These tools are set up using the API of OpenAI or other similar LLMs. To state this simply, LLMs exist in the form of an API, which can be metaphorically described as the central nervous system (CNS). The commercial tools mentioned above use this API/CNS to automate the systematic review process. In contrast, OpenAI allows the user to interact directly with the API/CNS. This is achieved by a specialized tool that converts human text into a language the API/CNS can process. This tool is used by the interface of ChatGPT.

It should therefore be clear that a direct comparison cannot be made between the commercial tools and the ChatGPT interface. Our group have used almost all of the major commercial tools available and have a clear picture of the advantages of the process described in this tutorial. However, providing a direct comparison would be beyond the scope of this report; perhaps even a separate report of its own.

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## Performing a systematic review using ChatGPT

Having a solid understanding of the above theoretical concepts on how ChatGPT works, provides a solid background for leveraging its capabilities. It should be noted that background knowledge on how systematic reviews are conducted would be required. It is therefore recommended that the reader is familiar with the PRISMA guidelines as a bare minimum.[6] In fact, the methodology described below is based on the PICOS framework and should therefore be considered applicable to study types that can be appropriately captured by these elements.

## Formulating the question: the north star of systematic reviews

At the heart of every systematic review is a well-formulated research question. This question provides direction, scope, and specificity to the review process.

Utilizing ChatGPT-4.0: cardiologists can employ ChatGPT to assist in refining their research questions. By presenting preliminary questions to the model, they can receive feedback, alternative viewpoints, and even identify gaps they may have overlooked. The iterative querying feature proves especially valuable here, enabling a continuous refinement process.

### Determining the PICOS: the pillars of a comprehensive question

An effective research question in a systematic review is typically structured using the PICOS framework:

- P - population/patient: refers to the specific group or cohort under investigation;
- I - intervention: this can be a treatment, procedure, or any change introduced to the population;
- C - comparator/control: represents the alternative against which the intervention is compared, often the standard treatment or a placebo;
- O - outcome: refers to the effects or endpoints being measured;
- S - study design: identifies the type of study or studies the review will encompass, *e.g.*, randomized control trials, cohort studies.

*Relevance to cardiology*

In cardiology, a typical PICOS question might be: "In patients with acute myocardial infarction (Population), how effective is drug X (Intervention) compared to drug Y (Comparator) in reducing mortality rates (Outcome) based on randomized control trials (Study Design)?"

Utilizing ChatGPT-4.0: cardiologists can leverage ChatGPT to explore potential PICOS components. For instance, if unsure about which outcomes to measure, the model can provide insights into commonly assessed outcomes in similar studies.

### Importance of formulating the question and determining PICOS in systematic reviews

- Direction and focus: they offer a roadmap for the review, ensuring that the process remains targeted and efficient;
- inclusivity and exclusivity: they define what will be included in the review and, just as crucially, what will be excluded;
- reproducibility: a well-defined research question and PICOS framework make the review reproducible, a cornerstone of scientific research;
- efficiency: they streamline the process, ensuring that resources, including time, are optimally utilized.

### Conclusion: the imperative of precision

In systematic reviews, particularly within the nuanced field of cardiology, precision is paramount. Formulating a clear research question and diligently determining the PICOS components sets the stage for a comprehensive, accurate, and insightful review. When combined with the capabilities of ChatGPT-4.0, cardiologists are empowered to navigate the vast seas of literature with a clear compass, ensuring that their systematic reviews truly stand as pillars of evidence-based medicine.

## Constructing targeted pubmed search strings with ChatGPT-4.0: delving into iteration and refinement

For the purposes of this tutorial, we will use two sample questions:

- investigating the effect of SGLT2 inhibi-

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tors on cardiac arrhythmias. This question could yield a significant number of results given the widespread use and interest in SGLT2 inhibitors, especially when considering both diabetic and heart failure populations;

• determining the optimal antiplatelet therapy for left ventricular thrombus post-myocardial infarction. This is a more specialized query and is expected to retrieve fewer, more focused results.

### The value of refining research questions

A preliminary research question often possesses the basic elements of what a researcher aims to uncover. However, to ensure the search is both comprehensive and precise, refining this question is essential. By narrowing down or expanding specific aspects of the question, you can better target the literature you aim to retrieve. In our examples, determining the specific patient population (diabetics, heart failure patients, or both) for the SGLT2 question will significantly impact the scope and volume of search results.

### Crafting and iteratively refining the search string

Strategies for crafting PubMed search strings:

• use of wildcards: using symbols like $ can be valuable to include variations of a term;

• MeSH (Medical Subject Headings): Implementing MeSH terms can focus the search using PubMed's curated vocabulary. But a word of caution: not all terms suggested by ChatGPT might correspond to actual MeSH terms;

• Boolean operators: utilizing "AND", "OR", "NOT" to provide structure to the search.

### The imperative of iteration in refinement

Engaging in an iterative process with ChatGPT aids in continual optimization. Here is how:

• feedback loop: after each search attempt, assess the results. If key articles are missing or if there's an overload of irrelevant ones, it is a clear signal to adjust your search string;

• adjustment with ChatGPT: with feedback in hand, ask ChatGPT for modifications. For instance, if the results for the SGLT2 inhibitors question are too broad, you might ask for a more narrowed-down string focused only on randomized controlled trials in diabetic patients;

• testing for MeSH validity: it is important to recognize that while ChatGPT is robust, it might sometimes suggest MeSH terms that do not exist. Always cross-check and verify. If a term is not recognized in PubMed, return to ChatGPT for a revision.

### Conclusions

Constructing a precise search string for systematic reviews is a balance of art and science. The combination of the PICOS framework, ChatGPT's iterative querying capabilities, and due diligence in verifying terms forms a powerful amalgamation. This ensures that the literature retrieval process is not only thorough but also sharply attuned to the research question's essence.

## Generating a PubMed search string

Let us try to use ChatGPT to generate a PubMed search string that addresses the two sample questions above. The process described below was executed on October 2nd, 2023.

### Question 1

*Prompt*

Please generate a PubMed search string that addresses the question: What is the effect of SGLT2 inhibitors on cardiac arrhythmias?

*Response*

Sure! To create an effective PubMed search string for the question, we can break it down into the major concepts:

• SGLT2 inhibitors;

• Cardiac arrhythmias.

To cast a wide net and ensure you capture relevant articles, you might also consider including synonyms or related terms. Here is a proposed search string: ("SGLT2 inhibitors" OR "sodium-glucose co-transporter-2 inhibitors" OR dapagliflozin OR empagliflozin OR canagliflozin OR ertugliflozin OR sotagliflozin) AND ("cardiac arrhythmias" OR "arrhythmias" OR "atrial fibrillation" OR "ventricular tachycardia" OR "atrial flutter" OR "ventricular fibrillation" OR "cardiac dysrhythmias").

This search string covers:

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• common SGLT2 inhibitors by their generic names;

• broad terms related to cardiac arrhythmias as well as specific types of arrhythmias.

You can use this string as a starting point and modify it based on preliminary results or if you have specific sub-questions in mind.

*Comment*

Upon investigation of the search string, a few things become immediately evident. Firstly, not all SGLT2 inhibitors were included in the search string. Secondly, the search string did not include any MeSH terms or wildcards. Therefore, it is advisable to iterate a better response.

*Prompt*

Please regenerate your response. This time please include all the generic names of the SGLT2 inhibitor class. Also, please make sure to use a combination of key words and MeSH terms. Finally, please use wild cards to improve the search string.

*Response*

Absolutely! Combining key words with MeSH terms and employing wildcards can certainly enhance the specificity and breadth of your PubMed search.

Here is a refined search string:

• SGLT2 inhibitors: generic names: Dapagliflozin, Empagliflozin, Canagliflozin, Ertugliflozin, Sotagliflozin, Ipragliflozin, Tofogliflozin, Luseogliflozin;

• cardiac arrhythmias: specific arrhythmias and general terms: Atrial fibrillation, Ventricular tachycardia, Atrial flutter, Ventricular fibrillation, Cardiac dysrhythmias;

• search string: ((“SGLT2 inhibitors” OR “sodium-glucose co-transporter-2 inhibitors” OR dapagliflozin* OR empagliflozin* OR canagliflozin* OR ertugliflozin* OR sotagliflozin* OR ipragliflozin* OR tofogliflozin* OR luseogliflozin*) OR (“Sodium-Glucose Transporter 2” [Mesh])) AND ((“cardiac arrhythmias” OR arrhythmia* OR “atrial fibrillation” OR “ventricular tachycardia” OR “atrial flutter” OR “ventricular fibrillation” OR “cardiac dysrhythmias”) OR (“Arrhythmias, Cardiac” [Mesh])).

This search string now:

• uses the wildcard * to capture various endings of the main terms and capture any derivative words;

• incorporates MeSH terms, which can be particularly useful as they will capture articles tagged with those MeSH terms even if the exact phrase is not used in the article.

Keep in mind that PubMed’s search algorithm does automatically search for plurals and various verb tenses, so certain wildcards might be redundant but can be useful in other databases.

*Comment*

When running the search string on PubMed on 10/2/2023, only 264 results were produced. Therefore, it is advisable to further refine the search string by broadening the search strategy.

*Prompt*

Please generate a broader search string using the guidelines I have provided.

*Response*

Absolutely. Let us broaden the search to capture more potentially relevant literature while adhering to the use of key words, MeSH terms, wildcards, and including all the generic names of the SGLT2 inhibitor class.

• SGLT2 inhibitors: generic names: Dapagliflozin, Empagliflozin, Canagliflozin, Ertugliflozin, Sotagliflozin, Ipragliflozin, Tofogliflozin, Luseogliflozin;

• cardiac arrhythmias: expanded to include related terms and conditions;

• here’s a broader search string: ((“SGLT2” OR “sodium-glucose co-transporter-2” OR dapagliflozin* OR empagliflozin* OR canagliflozin* OR ertugliflozin* OR sotagliflozin* OR ipragliflozin* OR tofogliflozin* OR luseogliflozin*) OR (“Sodium-Glucose Transporter 2” [Mesh] OR “Sodium-Glucose Transport Proteins” [Mesh])) AND ((arrhythmia* OR “atrial*” OR “ventricular*” OR dysrhythmia* OR tachycardia* OR fibrillation* OR flutter*) OR (“Arrhythmias, Cardiac” [Mesh] OR “Heart Diseases” [Mesh])).

This search string:

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• broadens the terms related to arrhythmias to include any mention of atrial or ventricular conditions;

• uses the wildcard * more liberally to cast a wider net, allowing for more inclusive results;

• expands the MeSH term usage to include "Heart Diseases" to capture more general cardiac-related literature that may discuss the effects of SGLT2 inhibitors.

Remember, broadening the search will fetch more results, but it may also yield more irrelevant articles. It is a trade-off between specificity and sensitivity.

*Comment*

This search was run on 10/2/2023 and produced 2,290 results. This was deemed appropriate, and the search string was considered to be finalized. Further refinements could be performed depending on the specifics of the project.

**Question 2**

*Prompt*

Please generate a Pubmed search string that addresses the following question: What is the optimal antiplatelet therapy for left ventricular thrombus post-myocardial infarction?

*Response*

Sure! When constructing a PubMed search string, it is important to break down the main components of your research question and then find relevant key words and Medical Subject Headings (MeSH) terms. Here is a suggestion for your question: ("left ventricular thrombus"[MeSH Terms] OR ("left"[All Fields] AND "ventricular"[All Fields] AND "thrombus"[All Fields]) OR "left ventricular thrombus"[All Fields]) AND ("myocardial infarction"[MeSH Terms] OR ("myocardial"[All Fields] AND "infarction"[All Fields]) OR "myocardial infarction"[All Fields]) AND ("antiplatelet agents"[MeSH Terms] OR ("antiplatelet"[All Fields] AND "agents"[All Fields]) OR "antiplatelet agents"[All Fields] OR "antiplatelet therapy"[All Fields]) AND ("treatment outcome"[MeSH Terms] OR ("treatment"[All Fields] AND "outcome"[All Fields]) OR "treatment outcome"[All Fields] OR "optimal"[All Fields]).

This search string combines:

• the condition of interest: left ventricular thrombus post-myocardial infarction;

• the intervention: antiplatelet therapy;

• the outcome: treatment outcome or optimal.

It is essential to review and refine the search as needed, especially if there are too many or too few results, or if some results are not relevant to the research question. Additionally, consider other MeSH terms, keywords, and filters to enhance the specificity and sensitivity of the search.

Finally, always keep in mind that the quality and relevance of search results can vary, and it is essential to critically assess each study based on its methodology, sample size, and findings.

*Comment*

This search string combines keywords with MeSH terms and seems to be acceptable. However, when running it on PubMed, only 28 results were produced. Furthermore, the following message is produced: [Quoted phrases not found in phrase index: "left ventricular thrombus," "antiplatelet agents"[]. This is something that occurs quite frequently, since ChatGPT has a tendency to invent keywords and MeSH terms. Finally, wildcards were not used. Therefore the prompt needs to be refined.

*Prompt*

Please regenerate your response. This time let us try a broader search strategy that utilizes wildcards along with keywords and MeSH terms. Also, when running your previous response on PubMed, I got the following message: Quoted phrases not found in phrase index: "left ventricular thrombus," "antiplatelet agents." Please avoid making this mistake.

*Response*

I apologize for the oversight in the previous search string. Here is a broader search strategy for PubMed, utilizing wildcards and combining both key words and MeSH terms: ("left ventricular" AND (thrombus* OR thrombo*)) AND ("myocardial infarction"[MeSH Terms] OR myocardial infarct*) AND (antiplatelet* OR "platelet aggregation inhibitors"[MeSH Terms])

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AND (optimal OR "treatment outcome"[MeSH Terms]).

Explanation:

• condition of interest: the phrase "left ventricular" is combined with the wildcard search for 'thrombus' or 'thrombo' to ensure a broader search related to left ventricular thrombi;

• associated condition: "myocardial infarction"[MeSH Terms] will capture the MeSH-indexed articles on myocardial infarction. The addition of the key word 'myocardial infarct*' with a wildcard will help fetch more articles that might not be strictly MeSH-indexed;

• intervention: the key word 'antiplatelet*' with a wildcard is used to catch various forms like 'antiplatelet therapy' or 'antiplatelet treatment'. Additionally, the MeSH term "platelet aggregation inhibitors" is included to be more comprehensive;

• outcome: the key word 'optimal' will help in finding the best or most favorable treatment, and "treatment outcome"[MeSH Terms] will aid in capturing articles related to the results of the antiplatelet therapy in this context.

When running the search string on PubMed, make sure to adjust for any specific requirements or limitations based on the initial results to refine and obtain the most relevant articles.

*Comment*

This search string produced 133 results without any error messages. This string was deemed appropriate.

## Screening by title and abstract

Once the search string has been generated and the search performed, the next step of the systematic review process is to screen the results by title and abstract. This process requires that the PICOS elements are well defined. Once that is defined, ChatGPT can be prompted to perform the screening. Let us use the example above with the effect of SGLT2 inhibitors on cardiac arrhythmias. In this case, the PICOS elements that correspond to the question under investigation would be:

• population (P): individuals with conditions for which SGLT2 inhibitors are prescribed, such as type 2 diabetes. This could also include patients with or without a history of cardiac arrhythmias;

• intervention (I): administration of SGLT2 inhibitors (*e.g.*, empagliflozin, canagliflozin, dapagliflozin);

• comparison (C): patients not receiving SGLT2 inhibitors, receiving a placebo, or receiving other antidiabetic medications;

• outcomes (O):

• incidence or prevalence of cardiac arrhythmias (this could be further specified, such as atrial fibrillation, ventricular tachycardia, etc.);

• any changes in the frequency or severity of existing arrhythmias;

• other related cardiovascular outcomes (*e.g.*, heart failure hospitalizations, cardiovascular death) that could indirectly hint at arrhythmic events;

• study design (S): this could vary based on the specific nature of the inquiry. Examples might include randomized controlled trials (RCTs), cohort studies, case-control studies, or systematic reviews.

In this particular case, we are interested in any patients that received SGLT2 inhibitors, without distinguishing between diabetics and heart failure patients. Therefore, the population element does not have to be part of our screening. The intervention will be an SGLT2 inhibitor. In terms of a control, we have no preference of placebo or active controls. Therefore, this step can also be excluded from our screening. The outcome of interest is any type of arrhythmia. Finally, we have also not specified a specific study design and can therefore exclude this part form out screening process. Therefore, the prompt for ChatGPT to perform the screening would be as follows.

*Prompt*

I am conducting a systematic review that investigates the effect of SGLT2 inhibitors on cardiac arrhythmias. I have run my Pubmed search and would like you to assist with screening by title and abstract. Please analyze the provided text and respond to the following questions:

• did the study investigate SGLT2 inhibitors, including canagliflozin, dapagliflozin,

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empagliflozin, ertugliflozin, sotagliflozin, ipragliflozin, tofogliflozin, or luseogliflozin?;

• did the study investigate cardiac arrhythmias, such as atrial fibrillation (afib), atrial flutter, supraventricular tachycardia (SVT), atrioventricular nodal reentrant tachycardia (AVNRT), atrioventricular reciprocating tachycardia (AVRT), atrial tachycardia, ventricular tachycardia (VT), ventricular fibrillation (VFIB), torsades de pointes, bradyarrhythmias, sinus bradycardia, atrioventricular block (first-degree, second-degree, and third-degree or complete heart block), sick sinus syndrome, premature atrial contractions (PACS), premature ventricular contractions (PVCS), idioventricular rhythm, junctional rhythm, wolff-parkinson-white syndrome (WPW), or long qt syndrome?

Please provide a "Yes" or "No" answer for these questions, along with your justification for that decision. If the provided text does not provide sufficient information for a decision to be made, please respond "Not Determined."

*Comment*

There are three main concepts that require further elaboration. Firstly, the questions ChatGPT is asked to generate should always be close ended or "Yes" or "No" type questions. It is also possible that the information in a title and abstract does not provide enough information for a decision to be made. This should always be accounted for in the prompt. Secondly, while ChatGPT knows which molecules fall within the SGLT2 inhibitor category, we have named them in the question. The reason for this is to save ChatGPT the computational strain of having to retrieve them. The different types of cardiac arrhythmias in the second question have been defined for the same reason. Ultimately, the information in the questions should provide as much detail as possible. This in turn will ensure that ChatGPT can focus on responding to the task at hand, rather than spending resources to figure out the question. Thirdly, we have chosen to eliminate some of the PICOS elements in this particular case. However, this may not always be valid. For example, we could be interested in diabetics on SGLT2 inhibitors, not heart failure patients. Alternatively, we could be interested in the exact opposite. In such cases we would add a third question:

• did the study include diabetic patients without heart failure?

The process is rather simple form that point forward. The title and abstract have to be inserted into ChatGPT along with the prompt. If ChatGPT produces a "Yes" for all questions (regardless of the number of questions), then the study will be included. If there is just one "No" answer, then the study does not meet our PICOS criteria and should therefore be excluded.

Let us try a specific example. For the purposes of this tutorial, we have asked ChatGPT to generate a fictional title and abstract for us to process using the screening prompt above.

*The effect of SGLT2 inhibitors on the incidence and severity of cardiac arrhythmias in type 2 diabetic patients*

BACKGROUND: This study sought to evaluate the impact of Sodium-Glucose Cotransporter 2 (SGLT2) inhibitors on the occurrence and severity of cardiac arrhythmias in patients diagnosed with type 2 diabetes mellitus (T2DM). Type 2 diabetes has been linked with a higher risk of cardiovascular diseases, including cardiac arrhythmias. There is growing interest in the potential benefits of SGLT2 inhibitors on cardiovascular outcomes, but their direct effect on cardiac arrhythmias has not been thoroughly explored.

METHODS: A total of 500 patients with T2DM were enrolled and randomized in a 1:1 ratio to receive either an SGLT2 inhibitor (dapagliflozin 10 mg daily) or a placebo. All patients were on standard diabetic care and continued their pre-existing treatments. The primary outcome measured was the incidence of any cardiac arrhythmia over a 12-month period, as detected by 24-hour Holter monitoring. Secondary outcomes included the severity and duration of arrhythmias, changes in left ventricular ejection fraction, and hospitalization due to arrhythmic events.

RESULTS: The SGLT2 inhibitor group demonstrated a significant reduction in the incidence of cardiac arrhythmias compared to the placebo group (12% *vs*. 24%; HR: 0.47; 95% CI: 0.32-0.69; P=0.001). The severity and duration of arrhythmias were also notably reduced in the treatment group. No significant difference in hospital-

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izations due to arrhythmic events was observed between the two groups. Left ventricular ejection fraction improved more in the SGLT2 inhibitor group compared to placebo, although this difference was not statistically significant.

CONCLUSIONS: The introduction of SGLT2 inhibitors in patients with T2DM demonstrated a potential protective effect against the development and severity of cardiac arrhythmias. Further studies on a larger scale and from multiple institutions are recommended to confirm these findings and explore potential mechanisms.

Now let Us process this title and abstract with the prompt we have generated.

*Response*

Did the study investigate SGLT2 inhibitors, including canagliflozin, dapagliflozin, empagliflozin, ertugliflozin, sotagliflozin, ipragliflozin, tofogliflozin, or luseogliflozin?

Yes. Justification: The study title and content clearly mention the investigation of SGLT2 inhibitors. Specifically, the text states that patients were randomized to receive either an SGLT2 inhibitor (dapagliflozin 10 mg daily) or a placebo.

Did the study investigate cardiac arrhythmias, such as atrial fibrillation (afib), atrial flutter, supraventricular tachycardia (SVT), atrioventricular nodal reentrant tachycardia (AVNRT), atrioventricular reciprocating tachycardia (AVRT), atrial tachycardia, ventricular tachycardia (VT), ventricular fibrillation (VFIB), torsades de pointes, bradyarrhythmias, sinus bradycardia, atrioventricular block (first-degree, second-degree, and third-degree or complete heart block), sick sinus syndrome, premature atrial contractions (PACS), premature ventricular contractions (PVCS), idioventricular rhythm, junctional rhythm, wolff-parkinson-white syndrome (WPW), or long qt syndrome?

Yes. Justification: The study's objective is to evaluate the impact of SGLT2 inhibitors on the occurrence and severity of cardiac arrhythmias in patients diagnosed with T2DM. Moreover, the primary outcome measured was the incidence of any cardiac arrhythmia over a 12-month period, as detected by 24-hour Holter monitoring. Even though the specific types of arrhythmias (*e.g.*, atrial fibrillation, atrial flutter) are not individually listed, the general term "cardiac arrhythmias" is repeatedly mentioned throughout the provided text, indicating that the study did investigate this outcome.

Therefore, this study should be included in the review.

Now, let us try a study that does not meet the criteria for inclusion. Here is another fictional title and abstract generated by ChatGPT.

*Efficacy of PCSK9 inhibitors versus statins in patients with atherosclerotic plaques: A comparative study*

BACKGROUND: Atherosclerotic plaques are major contributors to cardiovascular diseases and their sequelae, posing a significant threat to global health. With the advent of PCSK9 inhibitors, which target and inhibit proprotein convertase subtilisin/kexin type 9, there emerges a potential new avenue for the management of atherosclerotic diseases. This study aimed to compare the therapeutic outcomes and safety profiles of patients diagnosed with atherosclerotic plaques treated with either statins or PCSK9 inhibitors.

METHODS: We conducted a retrospective observational cohort study of patients diagnosed with atherosclerotic plaques from 1st January 2017 to 31st December 2022, who were treated with either statins or PCSK9 inhibitors. Demographics, medical histories, baseline investigations, and medication data were systematically recorded. The primary outcomes assessed were the progression of atherosclerotic plaques on imaging, cardiovascular events, and any drug-related adverse effects.

RESULTS: Of the 96 patients identified, 68 were managed with statins and 28 with PCSK9 inhibitors, including 17 on evolocumab and 11 on alirocumab. Most patients, 82 (85%), were male, with an average age of 60±15 years. Hypertension was identified as the primary risk factor in 79 (82%) of these patients. At baseline, both groups were comparably matched in terms of risk factors and plaque characteristics. Over a median follow-up of 2.7±1.2 years, there were no significant differences in plaque progression between the statin and PCSK9 groups (14% *vs.* 11%, P=0.56). However, cardiovascular events were marginally lower in the PCSK9 group (8%

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*vs*. 2%, P=0.09). Adverse events, primarily myalgias, were more pronounced in the statin group (18% *vs*. 3%, P=0.02).

CONCLUSIONS: Preliminary data indicate that PCSK9 inhibitors might present a viable alternative to statins in the management of atherosclerotic plaques, especially for those intolerant to statins due to side effects. While there was no significant reduction in plaque progression, the potential reduction in cardiovascular events and improved safety profile warrants further exploration. As always, definitive answers on the superior agent and indications for its use will be best elucidated by prospective, randomized trials.

Here is ChatGPT's response to the same prompt:

• did the study investigate SGLT2 inhibitors, including canagliflozin, dapagliflozin, empagliflozin, ertugliflozin, sotagliflozin, ipragliflozin, tofogliflozin, or luseogliflozin? Answer: No Justification: The study is focused on the efficacy of PCSK9 inhibitors *versus* statins in patients with atherosclerotic plaques. There is no mention of SGLT2 inhibitors in the provided text;

• did the study investigate cardiac arrhythmias, such as atrial fibrillation (afib), atrial flutter, supraventricular tachycardia (SVT), atrioventricular nodal reentrant tachycardia (AVNRT), atrioventricular reciprocating tachycardia (AVRT), atrial tachycardia, ventricular tachycardia (VT), ventricular fibrillation (VFIB), torsades de pointes, bradyarrhythmias, sinus bradycardia, atrioventricular block (first-degree, second-degree, and third-degree or complete heart block), sick sinus syndrome, premature atrial contractions (PACS), premature ventricular contractions (PVCS), idioventricular rhythm, junctional rhythm, Wolff-Parkinson-White Syndrome (WPW), or long qt syndrome? Answer: No Justification: The study is centered around the progression of atherosclerotic plaques and does not mention or focus on any of the cardiac arrhythmias listed.

Thus, this study does not seem relevant for the systematic review based on the criteria you've provided.

### Final notes

This process allows for efficient screening by title and abstract. ChatGPT seems to be very capable of making the correct decisions, provided it is prompted appropriately. A major limitation of this process is that it could actually be more time consuming than reading the title and abstract and determining its relevance (manual process). However, abstracts are usually limited to 250 words. This means that 5 or 6 abstracts can be processed at the same time in one interaction, considering the token limit. This would certainly produce a process that is effective and less time intensive than human screening. In these cases, the questions that ChatGPT is asked to respond to should be made as specific as possible to ensure computational capacity is preserved for the meaningful tasks.

The limitation of only being able to screen 5-6 abstracts at a time is certainly important. However, we cannot see any ways of bypassing that, considering the limited token window for each interaction. The only way that something like this can be overcome is by creating automation software that feeds the abstracts to ChatGPT automatically, without the requirement for human intervention. This can result in applications that are capable of screening hundreds or even thousands of titles and abstracts rapidly. In fact, we expect a few such tools to be made available soon.

## Generating study summaries for data extraction

The process of data extraction is perhaps the most resource intensive process of a systematic review. The size of the data that has to be processed is usually considerable and therefore may require more computational power than what is available to ChatGPT. Therefore, we propose the introduction of an intermediate step that involves the generation of a study summary. This way, ChatGPT can be instructed to analyze the results section of a report and generate a summary paragraph that contains only the information of interest. This summary paragraph can then be used to generate the data extraction tables. For example, let's say we are investigating the effect of a novel SGLT2 inhibitor on cardiac arrhythmias. Specifically, we are interested on the effect this agent had on the incidence of stroke as an indirect

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measure of the incidence of atrial fibrillation. For the purposes of this tutorial, we have asked ChatGPT to generate a fictional results section of a clinical trial that investigated the safety and efficacy of a novel SGLT2 inhibitor. We can then use the following prompt.

**Prompt**

Please analyze the provided text of a clinical trial that investigated the effect of a novel SGLT2 inhibitor on cardiac arrhythmias. Your task is to generate a summary paragraph that describes the study design, followed by a brief description of the patient baseline characteristics. Finally the summary paragraph should include results in terms of the incidence of stroke.

**Results**

*Trial overview*

Out of the 7342 patients with atrial fibrillation that participated, half (3671) were administered Luminabate, the novel SGLT2 inhibitor, and the other half (3671) received a placebo. The median follow-up period was 3 years.

*Baseline characteristics*

The baseline characteristics between the Luminabate and placebo groups were well-balanced. The average age of participants in the Luminabate group was 68.2 years (with a standard deviation of 8.5 years), while in the placebo group it was 67.9 years (with a standard deviation of 8.3 years). The gender distribution was slightly in favor of males, constituting 52.1% in the Luminabate group and 51.7% in the placebo group. The majority of the patients, about 78.5%, had a history of hypertension. The mean CHA2DS2-VASc Score, used for estimating the risk of stroke in patients with atrial fibrillation, was 3.2 in the Luminabate group and 3.3 in the placebo group.

*Primary outcome*

For the primary outcome, which is the incidence of complications, including stroke, there was a noticeable difference in favor of Luminabate. Over the 3-year follow-up period, the Luminabate group recorded a 7.5% incidence rate, compared to 10.8% in the placebo group. This translates to a relative risk reduction of 30.6%, which was found to be statistically significant with a P value of 0.002.

*Secondary outcomes*

The secondary outcomes also showed promising results for Luminabate. Hospitalization due to cardiac complications was lower for the Luminabate group at 12.3%, compared to the placebo group at 16.9%. This difference was statistically significant. Furthermore, overall mortality over the 3-year period was reduced to 5.1% in the Luminabate group, in contrast to 7.2% in the placebo group, again demonstrating a significant advantage for the Luminabate group.

*Subgroup analysis*

Diving deeper into specific subgroups of interest, Luminabate continued to showcase its efficacy. For patients aged above 75, 8.1% in the Luminabate group had complications, as opposed to 11.3% in the placebo group. Similarly, for patients with a CHA2DS2-VASc Score greater than 4, the Luminabate group had a 9.2% complication rate compared to 13.5% in the placebo group.

*Adverse events*

Regarding safety, 14.2% of patients in the Luminabate group reported adverse events compared to 13.7% in the placebo group. The most common side effects associated with Luminabate were urinary tract infections (4.6%) and genital fungal infections (3.5%). Dehydration was reported by 2.1% of the Luminabate group, and a minor 0.5% reported diabetic ketoacidosis. However, the overall serious adverse event rate remained comparable between the two groups.

**Response**

In a clinical trial investigating the effects of Luminabate, a novel SGLT2 inhibitor, on cardiac arrhythmias, 7342 patients with atrial fibrillation participated. They were evenly split, with half receiving Luminabate and the other half a placebo, monitored over a median period of 3 years. Base-

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line characteristics were consistent between both groups, with an average age of approximately 68 years, and a slight male predominance (52.1% in the Luminabate group and 51.7% in the placebo group). Roughly 78.5% of the patients had a history of hypertension. The mean CHA2DS2-VASc Score, indicating stroke risk, was 3.2 for the Luminabate group and 3.3 for the placebo group. When evaluating the primary outcome of stroke incidence, the Luminabate group showed a significant reduction, recording a 7.5% incidence rate over 3 years, in contrast to the 10.8% in the placebo group, resulting in a relative risk reduction of 30.6% (P value of 0.002). Secondary outcomes also favored Luminabate, including a decrease in hospitalizations and overall mortality. Adverse events were slightly higher in the Luminabate group, at 14.2%, compared to 13.7% in the placebo group, with urinary tract infections and genital fungal infections being the most prevalent side effects.

## Step 2

Now that the unnecessary text has been removed, we only have a paragraph that contains the information we are interested in. We can now prompt ChatGPT to generate a data extraction table.

### Prompt

Please analyze the provided text of a clinical trial that investigated the effect of a novel SGLT2 inhibitor on cardiac arrhythmias. Your task is to create a relevant table with the study design, the baseline characteristics, and the incidence of stroke.

In a clinical trial investigating the effects of Luminabate, a novel SGLT2 inhibitor, on cardiac arrhythmias, 7342 patients with atrial fibrillation participated. They were evenly split, with half receiving Luminabate and the other half a placebo, monitored over a median period of 3 years. Baseline characteristics were consistent between both groups, with an average age of approximately 68 years, and a slight male predominance (52.1% in the Luminabate group and 51.7% in the placebo group). Roughly 78.5% of the patients had a history of hypertension. The mean CHA2DS2-VASc Score, indicating stroke risk, was 3.2 for the Luminabate group and 3.3 for the placebo group. When evaluating the primary outcome of stroke incidence, the Luminabate group showed a significant reduction, recording a 7.5% incidence rate over 3 years, in contrast to the 10.8% in the placebo group, resulting in a relative risk reduction of 30.6% (P value of 0.002). Secondary outcomes also favored Luminabate, including a decrease in hospitalizations and overall mortality. Adverse events were slightly higher in the Luminabate group, at 14.2%, compared to 13.7% in the placebo group, with urinary tract infections and genital fungal infections being the most prevalent side effects.

### Response

Provided in Table I.

### Comments

In the generated table, the percentage of patients with hypertension was falsely generated. While ChatGPT warned that this percentage was "assumed, bot provided," it is still an inappropriate

TABLE I.—*Response.*

| Parameter / group | Luminabate group | Placebo group |
|---|---|---|
| Study design | | |
| Total N. of participants | 3671 | 3671 |
| Treatment | Luminabate | Placebo |
| Median follow-up period | 3 years | 3 years |
| Baseline characteristics | | |
| Average age | 68.2 years | 67.9 years |
| Male predominance | 52.1% | 51.7% |
| History of hypertension | 78.5% | 78.5% (assumed, not provided) |
| Mean CHA2DS2-VASc Score | 3.2 | 3.3 |
| Primary outcome: incidence of stroke | | |
| Incidence over 3 years | 7.5% | 10.8% |
| Relative risk reduction | 30.6% (P=0.002) | - |

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response. This is a clear indication that every single output has to be thoroughly examined. Beyond that, depending on the topic of investigation, we can instruct ChatGPT to include the exact data we are interested in. Furthermore, since we are performing a systematic review, this study will likely be one of many. Therefore the format of the table has to be slightly adjusted.

**Prompt**

Please regenerate your previous response. This time please use the rows as columns and vice versa.

**Response**

Provided in Table II.

**Comment**

This table provides a good starting point for the subsequent data extraction from other studies. Again, it cannot be stressed enough that every single cell that ChatGPT produces should be verified for accuracy. From our experience so far, the possibility of hallucinations is increased in the data extraction process. The intermediate step that we have introduced with the study summaries can significantly reduce these events. Nevertheless, it is crucial we double check that all the data has been transferred appropriately.

## Risk of bias assessment

The risk of bias assessment involves some preparation. Every study design has corresponding risk of bias assessment tools that have been validated. The method we use to perform a risk of bias assessment with ChatGPT involves the actual guidance document. These documents provide detailed instructions on how to perform an assessment. We can therefore use these instructions to ask ChatGPT to generate a relevant prompt. For the purposes of this tutorial, we will use the AMSTAR 2.0, the tool that is most commonly used to assess the methodological quality of a systematic review.[7] AMSTAR 2.0 has 16 domains. Here is the description of the first domain based on the guidance document:

Item 1: Did the research questions and inclusion criteria for the review include the components of PICO? It is common practice to use PICO description (population, intervention, control group and outcome) as an organizing framework for a study question. Sometimes timeframe should be added if this is critical in determining the likelihood of a study capturing relevant clinical outcomes (*e.g.* an effect of the intervention is only expected after several years). PICO identifies the elements that should be described in detail in the report of the systematic review and should enable the appraiser to judge selection of studies, and their combinability, and enable the user of the review to determine applicability of the results. Authors of systematic reviews do not always make the elements of PICO explicit but they should be discernable through a careful reading of the abstract, introduction and methods sections. To score 'Yes' appraisers should be confident that the 4 elements of PICO are described somewhere in the report.

**Prompt**

Please use the provided text that provides instructions on how to assess the first domain of AMSTAR 2.0. Your task is to generate a prompt that instructs ChatGPT to assess this domain.

**Response**

Please assess the first domain of AMSTAR 2.0 regarding the systematic review at hand. Specifi-

TABLE II.—*Response with rows and columns inverted.*

| Parameter | Total N. of participants | Treatment | Median follow-up period | Average age | Male predominance | History of hypertension | Mean CHA2DS2-VASc Score | Incidence over 3 years | Relative risk reduction |
|---|---|---|---|---|---|---|---|---|---|
| Luminabate group | 3671 | Luminabate | 3 years | 68.2 years | 52.1% | 78.5% | 3.2 | 7.5% | 30.6% (P=0.002) |
| Placebo group | 3671 | Placebo | 3 years | 67.9 years | 51.7% | 78.5% (assumed, not provided) | 3.3 | 10.8% | - |

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cally, evaluate whether the research questions and inclusion criteria for the review incorporate the components of PICO (Population, Intervention, Control group, and Outcome). If relevant, check if a timeframe is added, especially if it is crucial for determining the likelihood of capturing pertinent clinical outcomes, such as when an intervention's effect is anticipated only after several years. Ensure to look for descriptions of the four PICO elements within the report, even if they are not explicitly stated, by carefully examining the abstract, introduction, and methods sections of the review. Share your findings on whether the report sufficiently describes the PICO elements, and hence, whether it meets the criteria for this domain of AMSTAR 2.0. Clarify if the descriptions provided in the report allow for an appraisal of study selection and combinability, as well as the applicability of the review's results to a broader context.

### Comment

Supplementary Digital Material 1 (Supplementary Table I) provides a list of the prompts for all 16 domains of the AMSTAR 2.0. It should be noted that a similar process may be followed to generate prompts for other risk of bias assessment tools (*i.e.*, ROB 2.0, Newcastle Ottawa, etc.) (Supplementary Table I).

## SAILR checklist for reporting the use of ChatGPT in systematic reviews

We have developed the Systematic Artificial Intelligence Literature Review (SAILR) checklist for systematic reviews using ChatGPT and the methodology described in this tutorial. For the purposes of transparency, we would highly recommend that authors who utilize our methodology include this checklist. The SAILR checklist serves as a comprehensive guide to ensure rigor, transparency, and reproducibility in the research process. It outlines key steps from the formulation of research questions to the final synthesis and analysis, emphasizing the importance of detailed documentation at each stage. This includes the use of specific prompts in ChatGPT for data extraction and risk of bias assessment, methods for handling potential inaccuracies, and the necessity of manual checks. The checklist not only aids researchers in systematically employing ChatGPT in their reviews but also provides a structured framework for future researchers to follow, ensuring consistent and reliable use of AI tools in academic research (Table III).

## Discussion

In this tutorial, we have gone over the process of performing a systematic literature review using ChatGPT. Figure 1 provides a summary of the process, Figure 2 provides an overview of the advantages ChatGPT can provide, while Figure 3 provides some tips on the use of ChatGPT in systematic reviews, while Table IV describes the strengths and weaknesses of the process. One of the most important things that have to be taken into consideration when trying to prompt ChatGPT to perform these tasks (or any other task

TABLE III.—*SAILR checklist for reporting the use of ChatGPT in systematic reviews.*

| Step | Details | Reporting use of ChatGPT |
|---|---|---|
| Formulation of research question | Was the research question refined with ChatGPT? | |
| Generation of search strings | Document ChatGPT prompts for search strings and refinements. | |
| Title and abstract screening | Detail ChatGPT prompts for screening, including PICOS criteria. | |
| Data extraction process | Outline ChatGPT use in data extraction and intermediate steps. | |
| Risk of bias assessment | List ChatGPT prompts for each risk of bias domain and integration methods. | |
| Handling of hallucinations | Describe methods to identify and address inaccuracies in ChatGPT responses. | |
| Final synthesis and analysis | Explain the use of ChatGPT output in final review synthesis. | |
| Limitations and manual checks | Acknowledge limitations and necessity for manual verification. | |
| ChatGPT version and interaction | Record ChatGPT version and interaction dates for reproducibility. | |
| Token limitations and constraints | Mention token limits and strategies to mitigate computational constraints. | |

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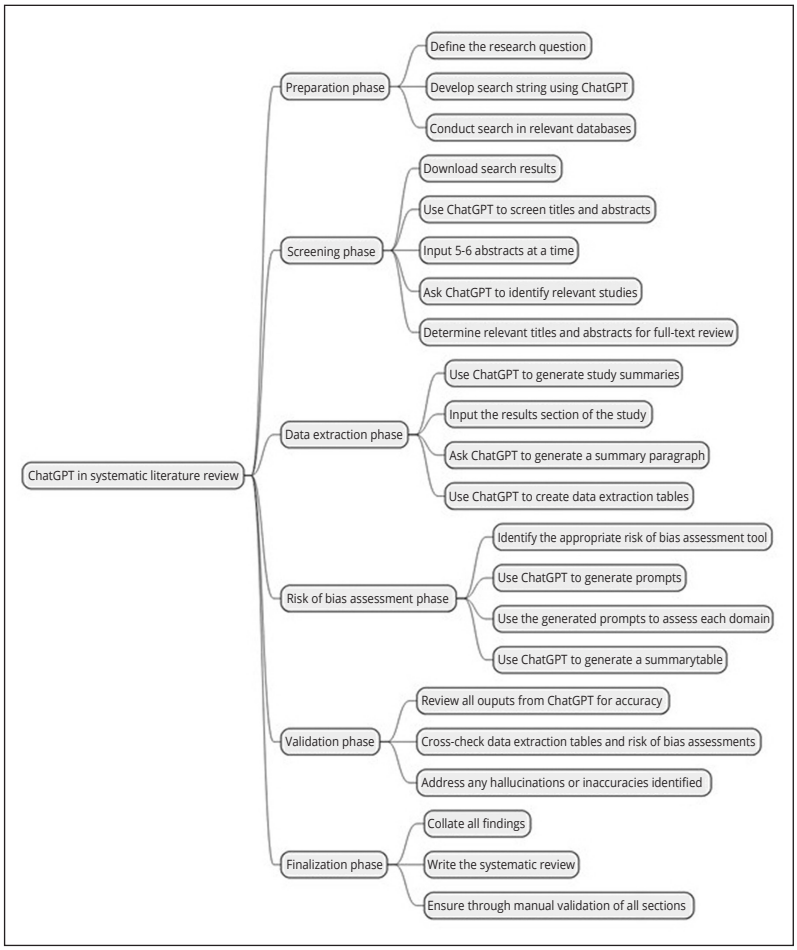



Figure 1.—Summary of the process of performing a systematic literature review using ChatGPT.

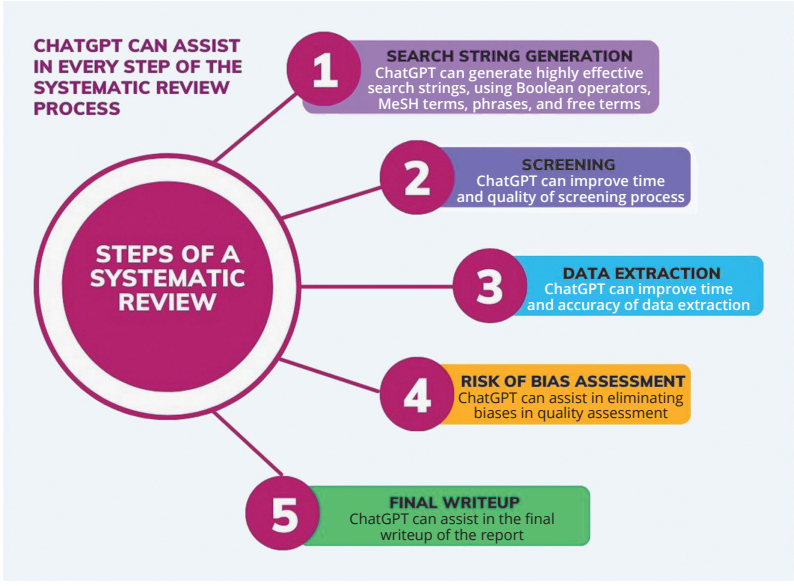



Figure 2.—Overview of the advantages ChatGPT can provide in systematic reviews.

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for that matter) is the limitations of the language model. Firstly, there is always the possibility of hallucinations. Hallucinations are defined as responses that the model has made up. Although the model has been noticeably improving in that area over the past few months, authors should be vigilant in double checking the output to make sure no such events have occurred. Ultimately, the authors are responsible for the validity of the information they publish. Therefore, every output must undergo extensive quality control to confirm its validity. As the user gains more experience with the model's behavior, the types of cases where hallucinations are more likely become evident. One good example of that is the word references *versus* the word citations. If the model is asked to retrieve references, there is a very high likelihood it will produce fake ones. If the word used is citations, hallucinations are much less frequent. While the reasons for this occurrence are beyond the scope of this tutorial, it is of the uttermost importance to gain experience with the model and learn its nooks and crannies. Another important limitation is the token (and therefore word) limit. Medical research often involves large pieces of text that cannot be processed reliably with one try. In these cases, the text has to be broken down into smaller batches and the subsequent output collated to construct a full response (Table IV).

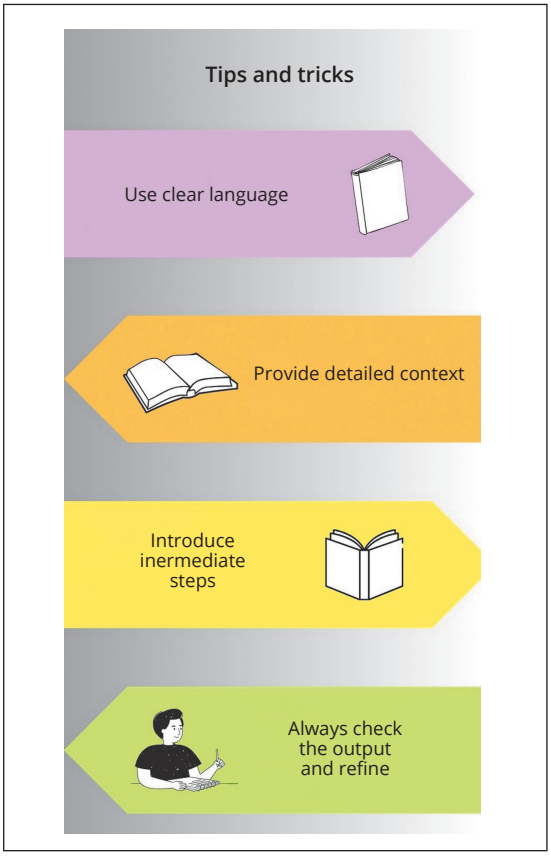



Figure 3.—Tips on the use of ChatGPT for systematic reviews.

The methods described in this tutorial have been tried successfully in other projects.[8] However, they are subject to continuous review and improvement. We have attempted to present an overview of our approach for every step of the systematic review. However, we do expect that these methods will be refined and improved substantially in the near future. We also expect that future versions of ChatGPT might allow for

TABLE IV.—*Strengths and limitations of using ChatGPT to assist in the systematic review process.*

| Aspect | Pros | Cons |
|---|---|---|
| Efficiency | Can process multiple abstracts simultaneously, potentially speeding up the screening process. | Token limit only allows processing of 5-6 abstracts at a time. |
| Data extraction | Can generate study summaries and data extraction tables from detailed results. | Possibility of hallucinations or inaccuracies in data extraction. |
| Risk of bias assessment | Can generate prompts based on guidance documents to assess risk of bias across multiple domains. | Requires detailed prompts and guidance documents for accurate assessment. |
| Token limit | Forces concise and specific prompts, which can lead to more focused interactions. | Large texts need to be broken down into smaller batches, potentially complicating the process. |
| Learning curve | Users can develop an understanding of the model's behavior over time, optimizing interactions. | Requires experience to understand the model's nuances and potential pitfalls (*e.g.*, word choice affecting responses). |
| General capabilities | Demonstrates potential in making correct decisions when prompted appropriately. | Always a possibility of generating hallucinated or inaccurate responses. |

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higher token limits. This will only serve to increase the amount of text that can be processed using our proposed methodology. Furthermore, we would advise against the use of the prompts we have provided as such. In fact, we would advise users to avoid using any type of pre-made prompts. There are two reasons for this. Firstly, there is no guarantee that the prompt will work to our satisfaction. More importantly, using prompts that were constructed by someone else takes away from building one's own style, developing an understanding of how to construct effective prompts, and ultimately being able to handle ChatGPT effectively.

The advent of AI capabilities through tools like ChatGPT indeed opens up a vast array of possibilities for conducting systematic reviews. As we have illustrated in this tutorial, there is a structured methodology that can be employed to leverage ChatGPT in enhancing the systematic review process. Yet, this methodology is in its nascent stages and there is a considerable scope for refinement. The collaborative efforts of researchers, over time, will undoubtedly contribute to honing these methods to be more efficient and effective. As more individuals immerse themselves in understanding the nuances of interacting with ChatGPT, we foresee a community of practice evolving, which will be instrumental in overcoming the challenges and limitations discussed earlier.

Moreover, the continuous evolution of ChatGPT and similar LLMs will naturally address some of the existing limitations. For instance, improvements in handling larger text inputs and reducing hallucinations are anticipated as these tools undergo further development. This, in conjunction with a growing user base gaining proficiency in employing these tools, sets the stage for significant advancements in the systematic review process.

The pathway towards automating systematic reviews using AI is filled with both challenges and opportunities. The ultimate aim is to significantly reduce the time and effort required to conduct systematic reviews, without compromising on the quality of the output. In fact, the goal should be to improve both speed and quality. This would not only expedite the clinical practice guideline development but also hasten the regulatory approval processes for new medical interventions, which is crucial for advancing healthcare outcomes.

As we venture into this uncharted territory, it is imperative that researchers are equipped with not just the technical know-how of utilizing AI tools like ChatGPT, but also a deep understanding of the iterative nature of this journey. Every project undertaken using these methods is a step towards refining the process, and sharing these experiences within the community is essential for collective advancement. Through a concerted effort, the goal of integrating AI seamlessly into the systematic review process is an achievable reality, poised to significantly contribute to the progress of medical science and public health (Supplementary Digital Material 2: Supplementary Text File 1).

## Key messages

- ChatGPT can be effectively used to assist in every step of the systematic review process.
- Authors should be extremely diligent in double-checking the quality of the output.

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*Conflicts of interest*

Giuseppe Biondi-Zoccai has consulted for Aleph, Amarin, Balmed, Cardionovum, Crannmedical, Endocore Lab, Eukon, Guidotti, Innovheart, Meditrial, Menarini, Microport, Opsens Medical, Terumo, and Translumina, outside the present work. All the other authors certify that there is no conflict of interest with any financial organization regarding the material discussed in the manuscript.

*Authors' contributions*

All authors read and approved the final version of the manuscript.

*History*

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*Supplementary data*

For supplementary materials, please see the HTML version of this article at www.minervamedica.it